



OPEN Biochemical evaluations of Ethiopian sweet sorghum [*Sorghum bicolor* (L.) Moench] accessions for sugar production

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Sweet sorghum [*Sorghum bicolor* (L.) Moench] is one of the varieties of sorghum having stalk with high concentration of soluble sugar. Biochemical evaluations of sweet sorghum enable it to determine the potential of the crop to be utilized for sugar production at industrial level. This study was designed to evaluate the Ethiopian sweet sorghum germplasms for production of crystalized sugar. In this study, 91 sweet sorghum accessions were evaluated based on total soluble sugar content. As a result, 8 accessions having high total soluble sugar scores were selected and evaluated for further biochemical quality traits. In addition, three agronomic parameters were used to evaluate their effect on the biochemical features of the studied sweet sorghum and sugarcane accessions. These traits include total soluble sugar (degree brix), polarization, sucrose content, sugar purity, days to maturity, stem height and stalk diameter. Three high quality sugarcane (*Saccharum officinarum*) genotypes collected from Kesseme sugar industry of Ethiopia were used as standard checks. Analysis of variance (ANOVA); mean separation and correlation were analyzed using R-software. ANOVA revealed significant variations for polarization value and purity of sweet sorghum juice at $p \leq 0.001$. The mean separation analysis revealed that maximum and minimum degree brix were obtained from T-11 (20.23%) and T-28 (16.88%), respectively. In the present study, the polarization values ranged from 41.57 (T-28) to 69.30 °Z (C-86/12). The three sugarcane standard checks showed relatively higher values of polarization in comparison with the sweet sorghum accessions. Among the sweet sorghum accessions, T-13 showed relatively higher polarization value whereas the remaining sweet sorghum accessions didn't show significant variation. The average sucrose content (pol percentage) was ranged from 11.98% (T-28) to 16.77% (C-86/12). C-86/12 sugarcane accession showed the highest (16.77%) mean value for sucrose content whereas T-28 sweet sorghum accession scored the lowest sucrose content (11.98%). Maximum purity of sugar was recorded from SP-70 (90.77%) sugarcane accession whereas; the minimum value was recorded from T-11 (63.62%) sweet sorghum accession. In addition, the correlation analysis revealed that there were both positive and negative correlation among biochemical and agro morphological traits of sweet sorghum and sugarcane. Brix value showed positive correlation with polarization (0.56**) and sucrose content (0.71**) while it was negatively correlated with sugar purity (-0.06), days to maturity (-0.06), stem height (-0.19) and stalk diameter (-0.16). On the other hand, polarization showed strong positive correlation with sucrose content (0.95**), purity (0.76**) and days to maturity (0.78**). In comparison with sugarcane, the studied sweet sorghum accessions revealed quite lower performance in polarization, sucrose content and purity. Nevertheless, this study confirmed the existence of a climate smart and promising sweet sorghum genotypes used for the production of sugar and syrup as an alternative sweetener.

Keywords Biochemical traits, Soluble sugar, Sweetener, Sweet sorghum accessions

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Sweet sorghum [*Sorghum bicolor* (L.) Moench] along with sugarcane and sugar beet is one of the varieties of sorghum having stalks with high sugar content^{1,2}. It is grown primarily for forage, silage, syrup production and as a source of renewable energy^{1,2}. It is the only crop that provides both grain and stem juice at a time^{3,4}. Research findings indicated that sweet sorghum is more preferable than sugarcane and sugar beet, due to its efficient photosynthetic mechanism, easy propagation from seed, low water and fertilizer requirements, short growth cycle (90 to 150 days), wide geographical adaptability and other related features^{1,5}.

Due to its high fermentable sugar content, global concerns over greenhouse gas emissions and an increasing demand for the production of ethanol, sweet sorghum is receiving significant global interest^{6–8}. Sweet sorghum has shown a potential to be a source of energy that could be an alternative for sugarcane to produce bio-fuel⁹. Furthermore, due to its tolerance to salt and drought, sweet sorghum is becoming more preferable than sugarcane and corn that are currently used for biofuel production in the world⁷.

Based on the type of sugar stored in its stem, sweet sorghum can be divided into saccharin- type sweet sorghum and syrup-type sweet sorghum^{7,10}. Saccharin-type sweet sorghum mainly contains sucrose and can be used for refining crystal sugar; whereas syrup-type sweet sorghum, which can be used for syrup production, contains mainly glucose^{7,11}.

For centuries, sucrose has been known and valued for its sweet taste. It is the most important component in the processing of sugarcane, resulting in a crystal form¹². Similar to sugarcane, sucrose accounts more than 75% of sweet sorghum juice¹³. Sucrose reaches its pick accumulation stage at physiological maturity of the crop¹³.

Historically in Ethiopia, sugarcane has been cultivated in farmers' fields since the 16th century and preceded commercial production^{14,15}. Currently sugarcane is produced by smallholder farmers in about 31,236.81 ha with 1,565,060 holdings in different parts of the country^{15,16} mainly for household consumption (chewing)¹⁵.

Sugarcane is the main sugar producing crop in Ethiopia and sugar industry plays a significant role in the socio economy of the country¹⁵. The history of the Ethiopian sugar industry goes back to the time when a center of sugarcane production was established in the Awash Valley during the Italian occupation in the 1930s¹⁷. The commercial sugarcane sector in Ethiopia started in 1951 with the establishment of the Wonji Sugar Factory, between Dutch company and the Ethiopian government^{15,17,18}.

The status of sugarcane plantations in different parts of the country in 2020 was estimated with area coverage of 105,000 hectares and production of 400,000 tons of sugar and 25,388m³ of ethanol per annum¹⁵. Currently, the average cane yield production from commercial varieties is around 114 t ha⁻¹^{15,19}. The Ethiopian sugar industry group planned to reach the production capacity of 726,670 tons of sugar by 2026/2027^{18,20}. To alleviate the current scarcity of sugar in the country, the Ethiopian government is working to increase sugar to 4.17 m t, and bio-ethanol production to 181 million liter^{15,19,21} through building modern sugar factories and expanding the existing ones²¹.

In Ethiopia, sugar demand is increasing from time to time; this is because of the increase in household consumption rates for sugar along with the expansion of the confectionery and soft drinks industries²². According to²³, sugar consumption is expected to increase due to country's population growth. In addition, the anticipated opening of new food and beverage factories in the country's planned agro-industrial parks is also expected to contribute to increasing demand for sugar in the coming years.

Since shortage of sugar is also a worldwide problem, currently research activities are being conducted to find alternative sources of table sugar such as natural sweeteners²⁴. One of the promising raw materials for their production of natural sweeteners is sweet sorghum^{5,8,24}.

In Ethiopia, sweet sorghum is grown along with grain sorghum for human consumption in rural areas of the country without any prior processing at industrial level. Despite the multipurpose importance of sweet sorghum worldwide in human nutrition and energy sector, Attention is not given for the Ethiopian sweet sorghum and utilizing it at higher industry level.

Sweet sorghum was identified as a source to produce syrup which can be used as a sugar alternative^{8,25}. The content of carbohydrates in the sweet sorghum juice is similar to honey^{24,25}. Historically, syrup obtained from the juice extracted from the stalk of sweet sorghum has been used as a sweetener in America since the 1890s and in India^{26,27}. In terms of nutrient content, sweet sorghum syrup is a better source of iron, zinc and calcium than honey^{13,28,29}. Since sweet sorghum syrup is rich in protein, essential amino acids and minerals²⁵, it has a tremendous potential for capturing a large market segment³.

Sweet sorghum juice has characteristics similar to sugarcane (*Saccharum officinarum*) juice and potentially used for sugar production³⁰. Since Ethiopia is the center of origin and diversity for sweet sorghum^{31–33}, it will be an ideal and promising raw material for production of sugar and natural sweeteners that help to address the actual problem with lack of sugar in the country. The studies of different scholars showed that sweet sorghum is a good alternative crop to sugar cane since it requires only one third of the amount of water that sugar cane needs and in the case of maturity period sweet sorghum matures in 100 to 120 days, while first sugar cane crop takes one year to mature. Similarly, on volume basis, sweet sorghum has higher sugar content compared to sugarcane³³.

Since sugarcane plantations need high irrigation facilities and infrastructure development, there is a need to find alternative sugar crops which are more economical. For the long term goal of finding alternative uses of sweet sorghum and substitute sugarcane in sugar industries, evaluating the natural potential of sweet sorghum for sugar production is required. Sugar producing industries need some standards in terms of biochemical composition and the quality of sugar. The percentage of sucrose, polarization, brix value and purity are widely used biochemical parameters to assess the quality of sugarcane juice. Biochemical evaluations of sweet sorghum enable it to determine the potential of the crop to be utilized for sugar production at industrial level. In this study, sweet sorghum accessions collected from different zonal administrations of Ethiopia were evaluated for their biochemical traits such as degree brix, polarization, sucrose content and purity percentage.

Materials and methods
Plant materials, plantation and experimental design

A total of 91 sweet sorghum accessions collected from North Shewa, South Tigray, North Wollo, South Wollo, East Gojam and Oromia Liyu Zone of Ethiopia were used in this study. These accessions were planted at the Akaki PolyTechnic College, Urban Agriculture farm which is found at the border of Oromia region and Addis Ababa City Administration of Ethiopia. It is located at 8° 86' 27 N" and 38°81'11"E. The study site has an altitude of 2120 m above sea level. The plantation was carried out from February to June in the off season of 2024 using irrigation. In addition, three commercial sugarcane varieties namely C-86/12, N-14 and SP-70 were collected from Kessem Sugar Industry and used as standard checks. The field experiment (plantation of the 91 sweet sorghum accessions) was laid out in alpha lattice design with three replications. The plot size was 2.25 m² consisting of equal number rows and blocks (10 rows and 10 blocks) with a spacing of 100, 75 cm and 50 cm between blocks, rows and plots respectively. Seeds were sown manually in rows at a seed rate of 10 kg/ha and thinned to adjust spacing between plants to 20 cm.

Screening of accessions for total soluble sugar (degree brix) content

When the sweet sorghum accessions reach at their physiological maturity (approximately 3 months), a juice sample was extracted from the middle internodes of stalk of all the 91 accessions and measured using digital Refractometer (RFM 960) to determine the total soluble sugar (degree brix). This was performed at the laboratory of School of Food and Biochemical Engineering, Addis Ababa University. Eight sweet sorghum accessions that had higher amount of total soluble sugar were selected for further sugar quality analysis and biochemical evaluations.

Determination of polarization, sucrose content and purity

Determination of polarization, sucrose content and purity of sweet sorghum juice were performed based on the methods and procedures recommended by^{34,35}. Eight sweet sorghum accessions that had higher total soluble sugar were taken to Kessem Sugar Industry for sugar biochemical evaluations. Approximately two kilogram of sweet sorghum stalk was chopped and juice samples were extracted using cutter grinder (CG03). The sweet sorghum juice was clarified by adding 2 g of 99% anhydrous basic lead acetate powder and 1 spatula of Kieselguhr in 100 ml of sweet sorghum juice and filtered through Whatman number 91 filter paper. To determine polarization values clarified juice was poured into a 200 mm polarimeter tube and read under Automatic saccharimeter (AUTOPOL 880). To determine pol percentage or sucrose content of the juice, pol reading score were converted in to pol percentage using SCHMITZ table. In addition, purity of the sweet sorghum juice was determined using the mathematical formula given as: $\text{purity} = \left(\frac{\text{pol\%}}{\text{degree brix}} \right) \times 100$

Data analysis

The biochemical data collected through laboratory tests were analyzed using R-software³⁶. In this study, analysis of variance (ANOVA) was utilized to find differences among accessions' means and to identify the significance of variation at $p \leq 0.001$ significance level. In addition, the least significant difference t-test (LSD) based on Fisher's LSD method (2010) at a probability of 0.05 was utilized to compare mean differences among accessions. In addition, to determine the strength and relationship among the different biochemical quality traits, a Pearson correlation analysis was carried out.

Results
Analysis of variance

In the present study, the analysis of variance (ANOVA) showed the presence of significant variations for polarization value and purity of sweet sorghum juice at $p \leq 0.001$. However, the total soluble sugar (degree brix) and sucrose content were recorded as non-significant biochemical traits (Table 1).

Analysis of mean separation and degree brix

Significant differences were not observed among accessions in terms of total soluble sugar (°Brix). Maximum degree brix was recorded for T-11 (20.23) followed by T-13 and C-88/12). On the contrary, the lowest degree

Traits	Source of variation	Df	Sum of square	Mean square	F value	Pr (> F)
Degree Brix	accession	10	37.72	3.77	0.481	0.884ns
	Residuals	22	172.41	7.84		
Pol reading	accession	10	2776	277.62	4.955	0.000849***
	Residuals	22	1233	56.03		
Sucrose content	accession	10	81.85	8.19	1.743	0.133ns
	Residuals	22	103.3	4.70		
Purity	accession	10	2631.2	263.12	84.69	34.1 × 10 ⁻¹⁵ ***
	Residuals	22	68.3	3.11		

Table 1. ANOVA for biochemical and quality traits. **Key:** ns = non-significant, significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Accessions	Brix value	Polarization (pol)	Sucrose content	Sugar purity
C-86/12	18.83 ^a	69.30 ^a	16.77 ^a	89.00 ^a
N-14	17.57 ^a	64.77 ^{ab}	15.74 ^{abc}	89.47 ^a
SP-70	17.56 ^a	65.58 ^{ab}	15.95 ^{ab}	90.77 ^a
T-11	20.23 ^a	50.66 ^c	12.82 ^{bcd}	63.62 ^c
T-12	18.18 ^a	46.99 ^c	13.31 ^{abcd}	72.83 ^b
T-13	20.17 ^a	53.13 ^{bc}	14.21 ^{abcd}	70.43 ^b
T-20	18.29 ^a	47.88 ^c	12.91 ^{bcd}	70.59 ^b
T-21	18.29 ^a	48.80 ^c	13.38 ^{abcd}	73.03 ^b
T-27	17.19 ^a	43.89 ^c	12.10 ^{cd}	70.27 ^b
T-28	16.88 ^a	41.57 ^c	11.98 ^d	70.63 ^b
T-84	17.40 ^a	44.83 ^c	12.55 ^{bcd}	72.07 ^b

Table 2. Mean values of brix, polarization, sucrose content, and sugar purity at 95% confidence interval. **Key:** Values in the same column with the same superscript are not significantly different at 0.05% significant level.

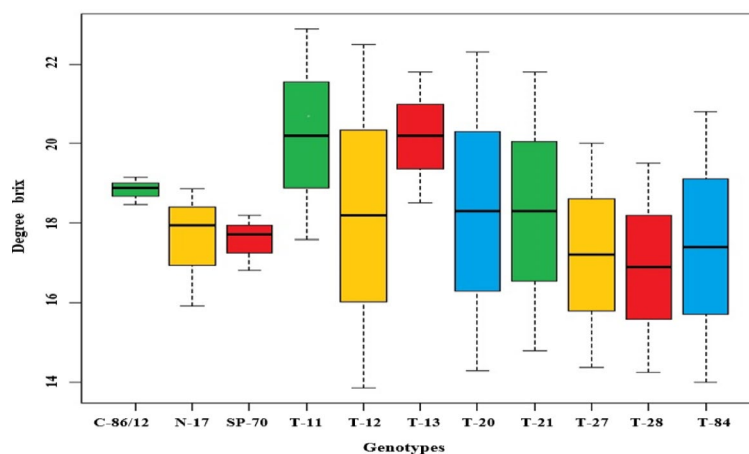


Fig. 1. Boxplot analyses of degree brix for the eight sweet sorghum and three sugarcane standard check accessions at 95% confidence interval, showing the distribution of degree brix trait for each accession. **Key:** The upper, median, and lower quartiles of boxes represent the 75th, 50th, and 25th percentiles of the clusters, respectively. The vertical lines in the graph indicate the variations observed in the accession and the dots show the outliers.

brix value was observed for T-28 (16.88) followed by T-27 and SP-70 (Table 2). As it is clearly shown in the box plot graph of LSD t-test for degree brix (Fig. 1), narrow interquartile range and short whisker were observed in the three sugarcane standard check genotypes. On the other hand, relatively fair level of interquartile range and whisker were observed among the eight sweet sorghum genotypes. Furthermore, visual comparison was depicted using the position of the median lines across the studied genotypes on the box plot graph (Fig. 1). The three sugarcane standard checks resulted in negative skewness while the eight sweet sorghum accessions showed an approximately symmetric skewness.

Analysis of polarization values

The polarization values ranged from 41.57 (T-28) to 69.30 °Z (C-86/12). In this case, significant differences have been observed. In the mean analysis of polarization most of the sweet sorghum accessions did not show statistical significant differences (Table 2). On the other hand, the three sugarcane accessions showed significant differences in all the eight sweet sorghum accessions. Among the eight sweet sorghum accessions, T-13 (53.13 °Z) showed relatively higher polarization value. The LSD t test for polarization (pol) values across the studied sweet sorghum accessions and sugarcane standard checks revealed variation in skewness, median position, interquartile range and whisker length (Fig. 2). Sweet sorghum genotypes T-13, T-28, T-11, T-84 and T-27 showed relatively short whisker length among the eight sweet sorghum genotypes. On the other hand, T-12, T-20, and T-21 showed long whisker length. In comparison with the three sugarcane standard checks, symmetric skewness was observed approximately in all sweet sorghum genotypes; whereas the three sugarcane standard checks with the exception of C-86/12 showed negative skewness (Fig. 2).

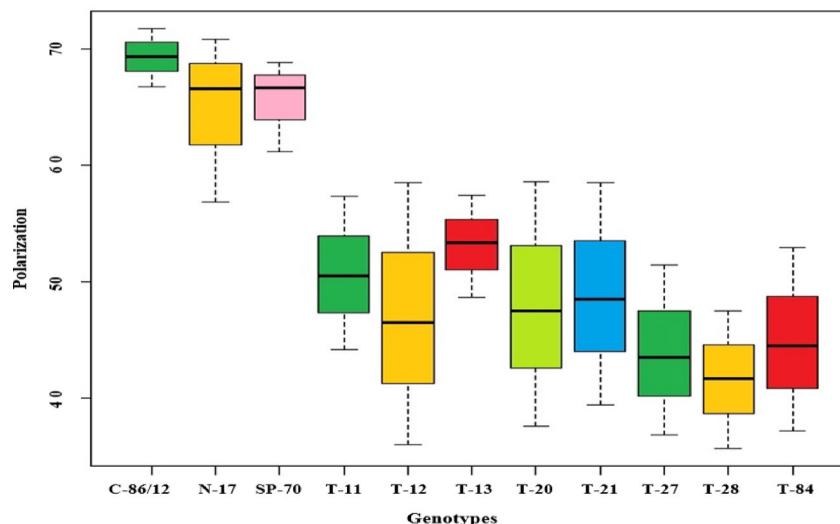


Fig. 2. Boxplot analyses of polarization for the eight sweet sorghum and three sugar cane standard check accessions at 95% confidence interval, showing the distribution of polarization trait for each accession. **Key:** The upper, median, and lower quartiles of boxes represent the 75th, 50th, and 25th percentiles of the clusters, respectively. The vertical lines in the graph indicate the variations observed in the accession and the dots show the outliers.

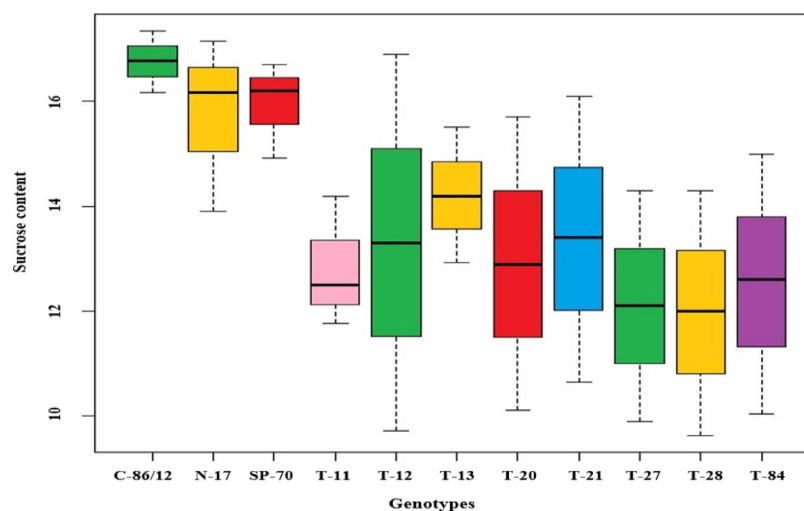


Fig. 3. Boxplot analyses of sucrose content for the eight sweet sorghum and three sugar cane standard checks accessions at 95% confidence interval, showing the distribution of sucrose content trait for each accession. **Key:** The upper, median, and lower quartiles of boxes represent the 75th, 50th, and 25th percentiles of the clusters, respectively. The vertical lines in the graph indicate the variations observed in the accession and the dots show the outliers.

Analysis of sucrose content

The average sucrose content ranged 11.98% (T-28) to 16.77% (C-86/12). In this analysis, significant mean differences have been observed in all accessions of sweet sorghum and sugarcane. C-86/12 sugarcane accession showed the highest mean values of sucrose content (16.77%) whereas T-28 sweet sorghum accession exhibited the lowest (11.98%) sucrose content. Following the three standard checks, T-13 (14.21%), T-21(13.38%), and T-12 (13.31%) sweet sorghum accessions showed the highest sucrose content, respectively (Table 2). Similarly, LSD t test for sucrose content across the studied sweet sorghum and sugarcane standard checks showed variation on the box plot graph (Fig. 3). Among the studied eight sweet sorghum genotypes, T-11 was positively skewed; whereas the remaining seven genotypes were more or less symmetrically skewed. In terms of interquartile range, T-11 and T-13 showed relatively narrow range and short whisker length. The sugarcane standard check genotype (C-86/12) showed the shortest whisker length and narrowest interquartile range. Genotypes N-14 and SP-70 showed negative skewness whereas C-86/12 showed approximately symmetrical skewness (Fig. 3).

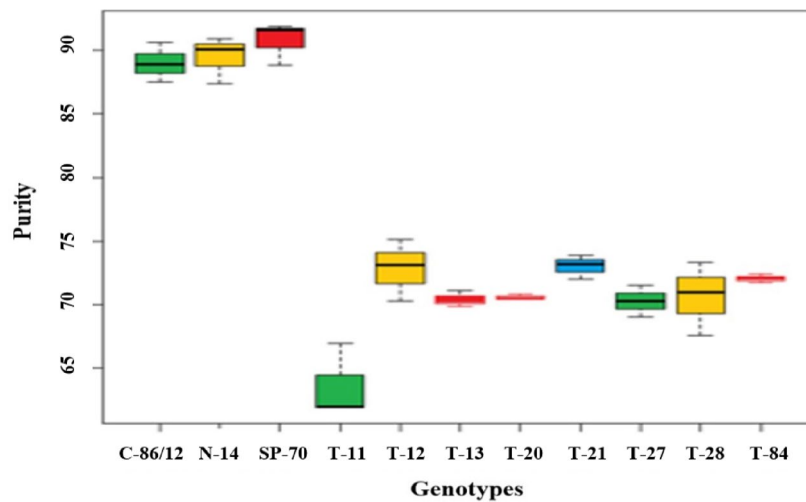


Fig. 4. Boxplot analyses of purity for the eight sweet sorghum and three sugarcane standard check accessions at 95% confidence interval, showing the distribution of purity trait for each accession. **Key:** The upper, median, and lower quartiles of boxes represent the 75th, 50th, and 25th percentiles of the clusters, respectively. The vertical lines in the graph indicate the variations observed in the accession and the dots show the outliers.

	Degree brix	Pol reading	Sucrose content	purity	Days to maturity	Stem height	Stalk diameter
Degree Brix	1						
Pol Reading	0.56**	1					
Sucrose Content	0.71* *	0.95**	1				
purity	−0.06	0.76**	0.65**	1			
Days to Maturity	−0.06	0.78**	0.62**	0.96**	1		
Stem Height	−0.19	−0.30	−0.22	−0.13	−0.26	1	
Stalk Diameter	−0.16	−0.51**	−0.44*	−0.48**	−0.50**	0.32*	1

Table 3. Correlation among biochemical quality and agro-morphological traits of sweet sorghum accessions and sugarcane standard checks. **Key:** ns = non-significant, significance codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘.’ 1.

Analysis of sugar purity

The analysis of purity showed that significant differences have been observed among the three sugarcane standard checks and the eight sweet sorghum accessions. Maximum sugar purity (90.77%) was recorded for SP-70 sugarcane accession whereas T-11 sweet sorghum accession showed the lowest (63.62%) sugar purity (Table 2). Sugar purity among the eight sweet sorghum genotypes ranged from 63.62% (T-11) to 73.03% (T-21). All sweet sorghum accessions with the exception of T-11 showed significant difference of purity. The graphical representation of the LSD t test for purity of sugar showed different image from the other three parameters. Both the sugarcane standard checks and eight sweet sorghum genotypes revealed very short whisker length and narrow inter quartile range (Fig. 4). Sweet sorghum genotypes of T-20 and T-84 possessed extremely short whisker length and narrow interquartile range. Sweet sorghum genotypes of T-11, T-13 and T-20 showed positive skewness whereas T-12 and T-21 were negatively skewed. The remaining sweet sorghum genotypes showed more or less symmetric skewness (Fig. 4).

Correlation among biochemical and agro-morphological traits

The correlation analysis revealed that there were both positive and negative correlation among biochemical and agro morphological traits of sweet sorghum and sugarcane. Brix value showed positive correlation with polarization (pol reading) (0.56**) and sucrose content (0.71* *) while it was negatively correlated with sugar purity (−0.06), days to maturity (−0.06), stem height (−0.19) and stalk diameter (−0.16) (Table 3). On the other hand, polarization showed strong positive correlation with sucrose content (0.95**), purity (0.76**) and days to maturity (0.78**). Besides it showed negative correlation with stem height (−0.30) and stalk diameter (−0.51**) (Table 3). The agro morphological trait of stem height was negatively correlated with all the recorded biochemical and agro-morphological traits except stalk diameter which had a significant positive correlation value of 0.32*.

In the above table each variable is represented by a row and a column. It shows correlation coefficients that determine the linear relationship between pairs of variables. The number in each cell ranges from −1 to 1. The positive numbers shows positive correlations (when one variable increases, the other variable tends to increase)

and the negative numbers shows negative correlations (when one variable increases, the other variable tends to decrease). A number of 0 means there is no correlation between the two variables. When the number is close to 1 (or -1), it shows the correlation is strong.

Discussions

Total soluble sugar

In the context of this study, degree brix ($^{\circ}$ Brix) refers to the total soluble sugar content of the juice of sweet sorghum or sugarcane. One degree brix implies that there is 1 gram of sucrose in 100 g of solution. Due to the presence of other solutes in the juice, degree brix can over-estimate sugar's soluble solid percentage³⁷. In the analysis of total soluble sugar, two sweet sorghum accessions (T-11 and T-13) showed higher degree brix than the standard check of sugarcane accessions. This could be due to the high photosynthetic efficiency of these accessions which resulted in rapid conversion of CO_2 and H_2O into carbohydrates. Very small variation of degree brix was observed among the eight sweet sorghum genotypes. In the present study, the degree brix value ranged from 16.88 to 20.23.

This result is more or less similar with the study conducted by³¹ which has recorded a degree brix value ranges from 11.8 to 22.5 with a mean value of 17.7 from a collection of 181 sweet sorghum accessions. In addition^{11,38,39}, have obtained a degree brix value of 15.05 to 21.50, 16 to 22.7, and 16–23, respectively. As it is clearly indicated by^{40,41}, the main requirement for bio-ethanol production is a Brix of 16% and above. This implies the higher the sugar content in the stalks, the higher the potential ethanol yield. Therefore, the degree brix range recorded in the present study has proved the potential of these sweet sorghum genotypes for the production of syrup and environmentally friendly bio-ethanol.

Polarization (pol value)

This parameter determines the amount of sucrose in the mixture of sugars, because in it, only sucrose diverts the plane of polarized light^{42,43}. In the present study, most of the studied sweet sorghum did not show significant difference. The three standard checks of sugarcane accessions showed higher polarization value than the sweet sorghum accessions. The highest polarization value was obtained from C-86/12 sugarcane genotype whereas the lowest value was obtained from T-28 sweet sorghum accession. Among sweet sorghum accessions, T-13 accession showed the highest value of polarization. In the previous study conducted by²¹ on quality differences among the three Ethiopian plantation, white sugars from Metahara, Finchaa and Wonji sugar factories with a range of 99.73–99.79 degree of pol ($^{\circ}$ Z) was recorded. This is much higher than the degree of pol ($^{\circ}$ Z) values recorded from sweet sorghum in the present studies. As it is pointed by^{34,44} different factors play a vital role in determining the value of polarization in sugar. The low value of polarization could be because of the fluctuation in the processing of raw sugar and also the interaction between coloring compounds and the impurities. On the other hand, the higher polarization value indicates the presence of more sucrose contents.

Sucrose content and purity of sweet sorghum

These two parameters determine the potential of sweet sorghum or sugarcane juice to produce crystalized sugar. The sucrose percent of sugarcane or sweet sorghum is the actual sugar present in the juice of the cane. On the other hand, purity refers to the percentage of sucrose present in the total solid content of the juice. A higher purity shows the presence of higher sucrose content in juice of sugarcane or sweet sorghum. If the sucrose content and the purity of a sugarcane or sweet sorghum juice attained the minimum values of 16% and 85%, respectively, it can fit for harvesting. In the present study, all the eight sweet sorghum accessions revealed relatively lower values of sucrose content in comparison with the three sugarcane standard checks. However, based on the sucrose content recommended specification of other countries such as⁴⁵, the sucrose percent of cane must be in a range of 14–21%.

Thus, the sucrose percent result for sweet sorghum genotype T-13 which had 14.21% sucrose content was within the acceptable and recommended range. Therefore, it can be used as a raw material for crystalized sugar production at industrial level. Significant mean variations were observed in all sweet sorghum accessions in terms of sucrose content. On the contrary, all of the studied sweet sorghum accessions did not show significant mean variation in the case of purity. With the exception of T-11 sweet sorghum accession, all of the others scored greater than 70% purity. A comparative study conducted to evaluate the performance of eight newly developed sugarcane genotypes by⁴⁶, reported 70.01% to 84.08% range of sugar purity. In this study, among the studied sweet sorghum accessions T-13, T-21 and T-12 showed a comparable performance with sugarcane standard checks and the results reported by previous studies^{11,46,47}. This implies that the studied accessions have a promising potential to be used as an alternative row material for sugar production in sugar industries and syrup production that serves as a natural liquid sweetener. However the overall evaluation in the biochemical features of sweet sorghum, relatively quiet low performance was observed in comparison with the sugarcane standard checks. Many researchers indicated that this variation could be due to the influence of variations in crop variety or the genetic makeup of the crop, changes in the agro climatic conditions, and fluctuations in the processing procedures⁴⁸.

Correlation among biochemical and agro-morphological traits

The correlation analysis revealed that there were both positive and negative correlation among the recorded biochemical and agro-morphological traits of sweet sorghum and sugarcane. The strength of these correlations were non-significant (weak), significant (moderate) or strong significant (high). In the present study, brix value showed a strong positive correlation with polarization (0.56**) and sucrose content (0.71**) while it was negatively correlated with sugar purity (−0.06), days to maturity (−0.06), stem height (−0.19) and stalk diameter (−0.16) (Table 3). The negative correlation of brix content with purity is in conformity to the general

mathematical formula of purity. This shows that purity of sugar is inversely proportional to the degree brix content of the sugar. In addition the increment in days to maturity, stem height and stalk diameter doesn't increase the total amount of soluble solid of the sweet sorghum juice rather it will be reduced. On the contrary, purity of sugar showed a positive and strong correlation with pol reading, sucrose content and days to maturity. This positive correlation implies that increasing in days to maturity or sucrose concentration in the juice of sweet sorghum or sugarcane will improve the purity of the sugar yield. According to^{49,50} the strength of correlation can be classified in to very weak (< 0.20), weak ($0.20-0.39$), moderate ($0.40-0.59$), strong ($0.60-0.79$) and very strong (> 0.80). In the present study, the correlation value ranged from -0.51 to 0.95 . The correlation of sucrose content with polarization, days to maturity and purity recorded in the present study was very strong. On the other hand, degree brix had very weak correlation with purity, maturity, stem height and stalk diameter. The result of this study confirmed that the three agro-morphological traits (stem height, stalk diameter and days to maturity) recorded in this study had a positive and negative effect on biochemical features of the sweet sorghum and sugarcane. Therefore any breeding programs that focus on improving the biochemical traits of these crops should consider the effect of these and other agronomic traits.

Conclusions

Biochemical characterizations have a great role to identify varieties with desirable technological characteristics that meet industrial requirements to produce the intended product. In this study, among the eight sweet sorghum accessions evaluated for sugar production potential, most of them showed good performance. However, in comparison with the sugarcane genotypes currently used by Kessem sugar industry in Ethiopia, they showed quite lower performance in terms of sucrose content and sugar purity. This may lead to low competitiveness with sugarcane in crystalized sugar production at industry level because crystalized sugar production is highly determined by purity and sucrose content. In addition, this study showed the existence of variation among the studied sweet sorghum and sugarcane accessions in terms of biochemical and phenotypic traits. Furthermore the correlation analysis indicated that the agro-morphological traits had both an antagonistic and positive effect on biochemical features of the studied sweet sorghum and sugarcane accessions.

There are many factors for low performance of sweet sorghum in terms of sucrose content and sugar purity. These might have been due to variation in fertility and moisture status of soil, genetic traits of the sweet sorghum accessions, or climatic conditions of harvesting time or growth stage of the sweet sorghum accessions during harvesting could significantly affected the purity of sugar and accumulation of sucrose in the stalk of sweet sorghum accessions.

Most of the sweet sorghum accessions of the present study showed high level of total soluble sugar content; this indicates that they are ideal for the production of syrup which serves as a liquid natural sweetener. Based on the evaluation results for sugar production from sweet sorghum, it is recommendable to use these promising sweet sorghum accessions for syrup production. This makes sweet sorghum as an alternative source of sweetener and will play its own role to minimize sugar shortage in Ethiopia. This also helps to ensure the food security of the people at large. Above all, this study opens a new avenue to transform sweet sorghum from the status of chewing as snack to utilize it at large industrial scale.

This study, which was mainly concerned to evaluate the potential of Ethiopian sweet sorghum for utilizing it as an alternative sugar crop in sugar industry is a new insight in the case of Ethiopia. Since Ethiopia is the center of origin and diversity for sweet sorghum, it will be an ideal and potential site for future research and breeding program. For the future, this study serves as a reference and a base to conduct further studies using more accessions. For full utilization of sweet sorghum in sugar production of Ethiopia at industrial level, further studies such as crystallization test and other biochemical assessment should be done by using additional sweet sorghum accessions. In addition, the assessment of economic profitability of sweet sorghum in comparison with sugarcane is also an essential research activity.

Data availability

All data generated or analyzed during this study are included in this published article.

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

In this study, sweet sorghum accessions collected from different parts of Ethiopia were used. In Ethiopian context any consent or ethical approval is not required for studies which involve plant materials for academic research.

Additional information

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